
Natural Language Processing (Almost) from Scratch

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● Core Idea

one unified neural network that can perform *many NLP tasks* like:

- Parts of Speech Tagging
- Chunking
- Named Entity Recognition
- Semantic Role Labeling

● Why this model ?

Previously NLP systems relied on :

- Heavy feature engineering (manually crafted rules / features)
- Task specific designs (one model per task)

That made systems **complex** and **hard to generalize**.

● How ?

They used **neural networks architecture** that learns automatically from data, even from **unlabeled text** (raw text) like:

- Word Embeddings
- Minimal prior linguistic knowledge

They called it **Almost from scratch** because they start from raw text (no handcrafted linguistic knowledge), but still rely on basic preprocessing and labeled data.

● Benchmark Models

Task	Benchmark	Data set	Training set (#tokens)	Test set (#tokens)	(#tags)
POS	Toutanova et al. (2003)	WSJ	sections 0–18 (912,344)	sections 22–24 (129,654)	(45)
Chunking	CoNLL 2000	WSJ	sections 15–18 (211,727)	section 20 (47,377)	(42) (IOBES)
NER	CoNLL 2003	Reuters	“eng.train” (203,621)	“eng.testb” (46,435)	(17) (IOBES)
SRL	CoNLL 2005	WSJ	sections 2–21 (950,028)	section 23 + 3 Brown sections (63,843)	(186) (IOBES)

System	Accuracy
Shen et al. (2007)	97.33%
Toutanova et al. (2003)	97.24%
Giménez and Màrquez (2004)	97.16%

(a) POS

System	F1
Shen and Sarkar (2005)	95.23%
Sha and Pereira (2003)	94.29%
Kudo and Matsumoto (2001)	93.91%

(b) CHUNK

System	F1
Ando and Zhang (2005)	89.31%
Florian et al. (2003)	88.76%
Kudo and Matsumoto (2001)	88.31%

(c) NER

System	F1
Koomen et al. (2005)	77.92%
Pradhan et al. (2005)	77.30%
Haghighi et al. (2005)	77.04%

(d) SRL

● Evaluation

They followed **official CoNLL challenge protocols**

- They used validation sets to tune model parameters fairly

Official scoring tools

- Conllevl (POS, NER, Chunking)
- Srl-eval.pl (SRL)

● The Networks

Window Approach:

- The window approach tags apply to the word located in the center of the window

Sentence Approach:

- The tags in the sentence approach apply to the word designated by additional markers in the network input.

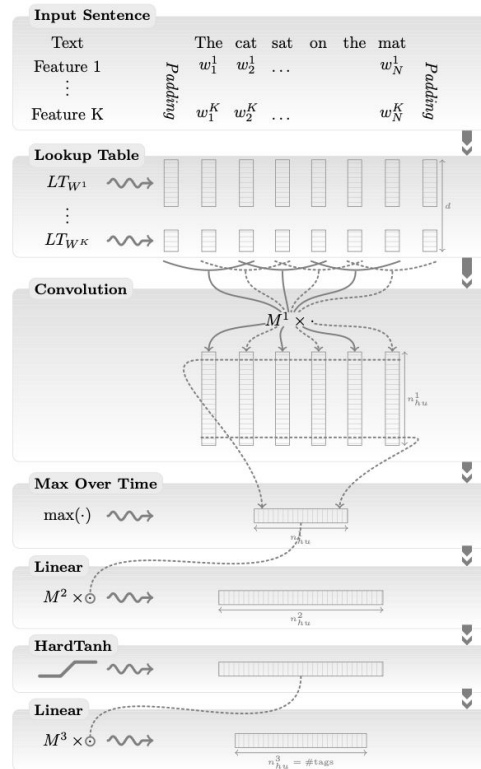
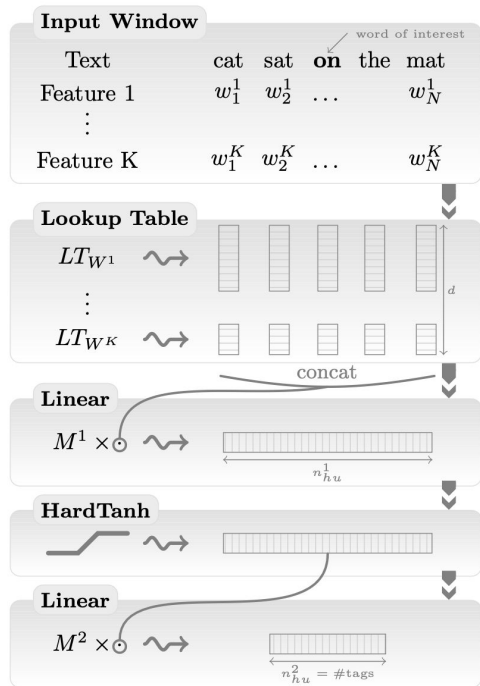
● Window Approach

- Assumes the tag of a word depends mainly on its neighboring words.
- The NN inputs a fixed size window of words.
- The window of words is passed through a lookup table layer and produces a matrix with dimensions of the word features by window size, this matrix gets concatenated to a column vector and passes through the rest of the network.
- The output layer size is equal to the number of possible tags for the task of interest, and can be interpreted as a score of the corresponding tag.

● Sentence Approach

- Improves on the window approach for the SRL task
- In this approach the complete sentence is passed through the lookup table layer.
- Local features around each word are produced by convolutional layers.
- These features are combined into a global feature vector which can be fed into standard affine layers.

● Window Approach / Sentence Approach



● Word Level -vs- Sentence Level Log Likelihood

Word-Level Log-Likelihood

- The model predicts each word's tag **independently**.
- Simple but ignores the **dependencies** between neighboring tags (e.g., “New” → B-LOC, “York” → I-LOC).
- Often produces **inconsistent tag sequences**.

Sentence-Level Log-Likelihood

- Considers the **entire sequence** of words and tags together.
- Adds **transition scores** between tags (e.g., $\text{tag}_1 \rightarrow \text{tag}_2$).
- Encourages **valid tag paths** across the whole sentence while **penalizing invalid ones**.
- Leads to smoother, more realistic predictions — similar in spirit to a **CRF layer**.

● Two-Stage Unsupervised Language Modeling

Model 1:

- Trained on English Wikipedia
- Gradually increased dictionary size (5 k $\rightarrow$ 100 k words)
- Ran $\approx$ 4 weeks $\rightarrow$ produced initial word embeddings

Model 2:

- Initialized with Model 1 embeddings
- Added Reuters RCV1 (+130 k vocabulary)
- Trained $\approx$ 3 additional weeks $\rightarrow$ final embeddings

Purpose: learn **robust word representations** from $\approx$ 852 million unlabeled words for later NLP tasks.

● Multi-Task Learning (MTL)

- Instead of training separate models for each task (POS, CHUNK, NER, SRL), **Multi-Task Learning (MTL)** involves training a single neural network to predict the labels for **all tasks** simultaneously.



Experimental Outcome

Task	Single-Task Performance (F1/PWA)	Multi-Task Performance (F1/PWA)	Improvement
Chunking (CHUNK)	93.63%	94.10%	Significant Gain
Named Entity Recognition (NER)	88.67%	88.84%	Small Gain
Part-of-Speech (POS)	97.20%	97.29%	Small Gain

● Strengths

- **No Feature Rules:** The computer **learned its own features**, making manual work (feature engineering) unnecessary.
- **Universal Model:** One simple design worked for all **four major NLP tasks**.
- **Lightning Fast:** The final system (**SENNA**) was incredibly **fast** and efficient.

● Limitations

- **Needed Structure:** The model still needed **Structured Decoding (SLL)**, to check for **tag patterns** across a sentence. It failed without **SLL**.
- **Not 100% from Scratch:** To get the absolute best score, they had to add a **tiny piece of basic outside information** (parse tags).

● Why This Paper Matters

- **The Embedding Revolution:** This paper proved that **Word Embeddings** are the most powerful feature in NLP.
- **Inspired the Future:** It directly led to key inventions like **Word2Vec** and established the method of **pre-training**, which is the core idea behind models like **BERT** and **GPT**.

● The SENNA Tool

- **SENNA** is the **free, high-performance tagging system** created from this research.
- **Benefit:** It offered competitive accuracy while being significantly **faster and lighter** than all older systems.
- **Final Impact:** SENNA gave the NLP world a practical, fast tool that proved the power of **deep learning word embeddings**.

● Conclusion – Three Essential Takeaways

- **Feature Learning > Feature Engineering:** Neural networks can learn features better than people can design them.
- **Unlabeled Data is key:** Learning general **Word Embeddings** from huge, unlabeled datasets is the key to overcoming the scarcity of expensive labeled data.
- **Generalization is Possible:** A single, **unified architecture** can successfully be applied to multiple, diverse NLP tasks.



Thank You !

Any questions? Suggestions?
Feedback?